

Anurag Mishra

Medium: anuragmishra_27746

Github: github.com/anurag-mishra899

Email: anurag.mishra899@gmail.com

Mobile: +91-9456243057

LinkedIn: in/anurag-mishra-660961b7

PROFILE SUMMARY

- **Experience:** 7+ YoE in designing and developing production-grade **Machine Learning, Deep Learning, Generative AI, and NLP** solutions for various industries, including retail, banking, telecommunications, and chemicals
- **Frameworks:** PyTorch, Langchain, Crewai, Autogen, LlamaIndex, Fastapi, Scikit, NLTK, SpaCy, TensorFlow, Keras
- **Tools:** Kubernetes, Docker, GIT, PostgreSQL, MySQL, SQLite
- **Platforms:** Azure Cloud (OpenAI), AWS (Claude Sonnet), GCP, Linux
- **Certified in:** Azure Fundamental, Azure Data Scientist, Databricks ML Professional
- **AI Conference as Speaker:** LLM Application, Agentic Workflow, Multi Modal LLMs
- **Blogs/Publications:** Article 1, Article 2, Article 3, Article 4

EXPERIENCE

- **EY GDS** Hybrid
Associate Manager (Full-time) Feb 2022 - Present
 - **GenAI Chatbot for 1500+ Bank Branches, Serving 9,000–10,000 Daily Users:**
 - * Led a team of 7 to design and deploy a GenAI chatbot for querying bank policies, circulars, and bulletins from SharePoint, reducing resolution time from **5–10 minutes to 30 seconds**.
 - * Developed a custom ingestion and chunking pipeline to process **6,000–7,000 complex PDFs** with tables and flowcharts, boosting retrieval performance by **55%**.
 - * Built a scalable backend using FastAPI, LangChain, Azure API Gateway, and Kubernetes with **region-based load balancing, throttling, and fallback**, cutting downtime by **50%**.
 - **Automated Post-Call Action Using Fine-Tuned SLMs for Contact Centers:**
 - * Fine-tuned SLMs using **LoRA techniques** on curated instruction datasets to generate structured post-call summaries aligned with business SOPs.
 - * Designed an end-to-end pipeline with **LangChain, Azure DevOps, ServiceNow APIs, and optimized inference (paged attention, KV caching etc)** on A100 GPUs for intelligent ticket handling.
 - * Enabled processing of **300–400 daily calls per center**, improving agent productivity by **60%** and reducing wrap-up time by **40%**, driving **\$2M annual cost savings**.
 - **Multi-Agent LLM System for Customer Support Triage and Query Resolution:**
 - * Designed specialized agents including **Intent Router, Information Synthesizer, Policy Validator, Escalation Negotiator**, each prompted for domain-specific functions and operating on shared memory via vector DBs
 - * Integrated with enterprise knowledge bases (SharePoint, Confluence) and CRM tools via **Azure Functions and Logic Apps**, enabling contextual awareness and real-time task hand-offs.
 - * Designed a comprehensive evaluation framework combining **task-level benchmarks, agent handoff accuracy, and dialogue coherence metrics (BLEU, ROUGE, GPT-4 evals)** to iteratively refine agent behaviors and ensure reliability
- **Deloitte (Office of US)** Remote
Consultant - AI (Full-time) Feb 2021 - Feb 2022
 - **Fine-tuning Transformer Models Reducing Operational Costs by 70%:**
 - * Delivered advanced techniques in deep learning model tuning, leading to significant improvements in **automation extraction workflows** across the organization; measured performance gains resulted in reduced processing times of critical data inputs by over 80% hours monthly.

- * Deployed deep learning models as **batch endpoints utilizing Azure Web Apps, Databricks Jobs, and MLflow**, ensuring streamlined integration and efficient model management, reducing operational costs by 70%.

- **Fixture Optimization for Inventory Management, Achieved \$5 Million Cost Reduction:** Devised optimization solution to allocate the optimal number of fixtures for each product and brand in retail chain stores, resulting in a \$5 million reduction in costs and expenses.
- **Sales Forecasting for Retail Client, Enhanced Demand Forecasting by 45%:** Developed sales forecasting model for each department per store to assess demand. Utilized tree-based algorithms such as Gradient Boosting, XGBoost, and LightGBM, as well as regression-based models like ARIMA & SARIMA.

• Virtusa Polaris

Hybrid

Machine Learning Engineer (Full-time)

June 2018 - Feb 2021

- **Multi-Class classification of Trading Products, Reducing Manual Tagging from 15k/hour to 1 Million/hour:** Devised solution for Product Controllers to reduce person-hours spent on tasks of identifying breach in trading activity.
- **NLP Models to Assess Commentary using SpaCy, Improving Comments Quality by 80% :** Developed NLP solution to identify the inconsistency in PnL commentary generated by Product Controllers. Milestones included building Rest API to render user request and ETL operations to handle user requests.
- **ML Models & Techniques:** Utilized various ML algorithms like Random Forest, Logistic regression Gradient Boosting and Hyper-parameter tuning to get 95% accuracy.
- **Tools & Technologies:** Used Python programming and worked on libraries like Pandas, Scikit-learn, NumPy, Matplotlib, Seaborn, Spacy

PROJECTS

- **Fine-Tuned LLM for Domain Specific tasks using LoRA, Increasing Domain-based NER By 55%:** Performed Fine-tuning of LLM models(LLaMA 3.1 8B, Mistral 7B) on 80 GB A-100 GPU on custom build dataset for Manufacturing domain.
- **LLM Based Agentic flow for Text to Code Solution:** Agent based solution that convert requirements to code. The solution uses multi-agents (debugger, coder, tester etc) to write the bug-free and optimized code.

EDUCATION

- **National Institute of Technology, Manipur** India
Bachelor of Technology - Information Technology; GPA: 8.5 *July 2014 - June 2018*
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases

HONORS AND AWARDS

- Awarded Excellence Client Performance Award in EY
- Honored to be among top 10 students in CSE Department
- Achieved (Among Top 1 % in board) in the entire district in HSC exam.